

Epidemiological Trends and Spatio-Temporal Analysis of Human Brucellosis and Associated Risk Factors in El Tarf, Northeast Algeria

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Introduction

- In Algeria, despite the implementation of control programs over several years, human brucellosis remains a significant public health concern.
- The incidence rate of human brucellosis remains high, recorded at **24.41 per 100,000 inhabitants in 2017** (INSP). This major zoonosis, listed among legally contagious diseases (OIE, 2017), has a significant social and economic impact on affected populations.
- This concerning epidemiological situation led us to conduct a retrospective descriptive study of human brucellosis in the El Tarf region.

Objectives

- Calculate the annual and monthly incidence of human brucellosis cases over a twelve-year period (2010–2021).
- Analyse the spatial and temporal evolution of human brucellosis in the study area.
- Identify risk factors associated with human brucellosis.
- Evaluate the performance of the epidemiological surveillance system for human brucellosis based on passive case detection data

Study Methodology

1- Study Design

Retrospective descriptive study conducted over twelve years (2010–2021) to analyze human brucellosis cases in El Tarf.

2. Data Collection

Human brucellosis case data obtained from the Health and Population Directorate of El Tarf.

Data categorized by **sex, age, region, month, and year** of diagnosis.

3. Serological Diagnosis

Diagnosis based on serological tests detecting antibodies:

- Rose Bengal test and Wright's serum agglutination test.

Blood samples were collected at diagnosis before treatment initiation.

4. Data Analysis and Spatial Mapping

- Data entered into Microsoft Excel and analyzed using SPSS v23.

- Calculated **annual incidence rates** as new cases per year divided by total population.

- Used **Kruskal-Wallis** and **Chi-square** tests to assess differences in disease trends by year, month, and season; significance set at $p < 0.05$.

- Created **choropleth maps** of case distribution with the online tool Datawrapper (<http://datawrapper.de/>) for spatial analysis.

Results

1- Annual Evolution

- From 2010 to 2021, **296 human brucellosis cases** were reported in El Tarf, averaging **24 cases/year**.
- Incidence increased significantly after 2014, from **4.47 to 9.38 cases per 100,000 inhabitants** in 2021 (more than double).
- The highest annual incidence was in **2019: 11.43 cases/100,000**.
- A significant difference was observed between years (Kruskal-Wallis, $p = 0.003$).

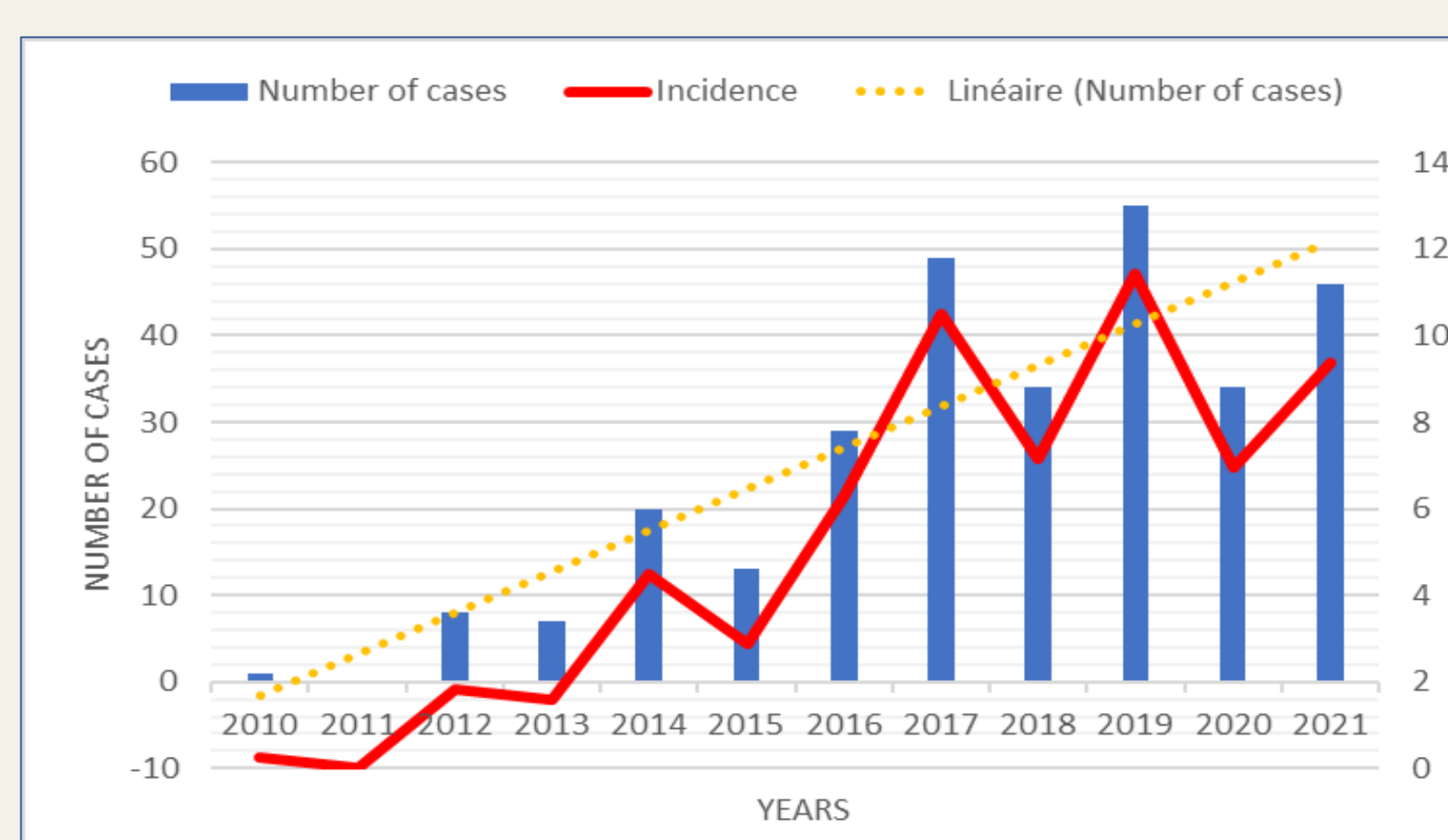


Figure 1: Evolution of Human Brucellosis Cases and Annual Incidence in El Tarf Province (2010–2021)

2- Temporal Distribution

- Cases occurred throughout the year with no significant monthly variation ($p = 1.00$).
- Peak months were April (35 cases), June (33 cases), and May (31 cases).

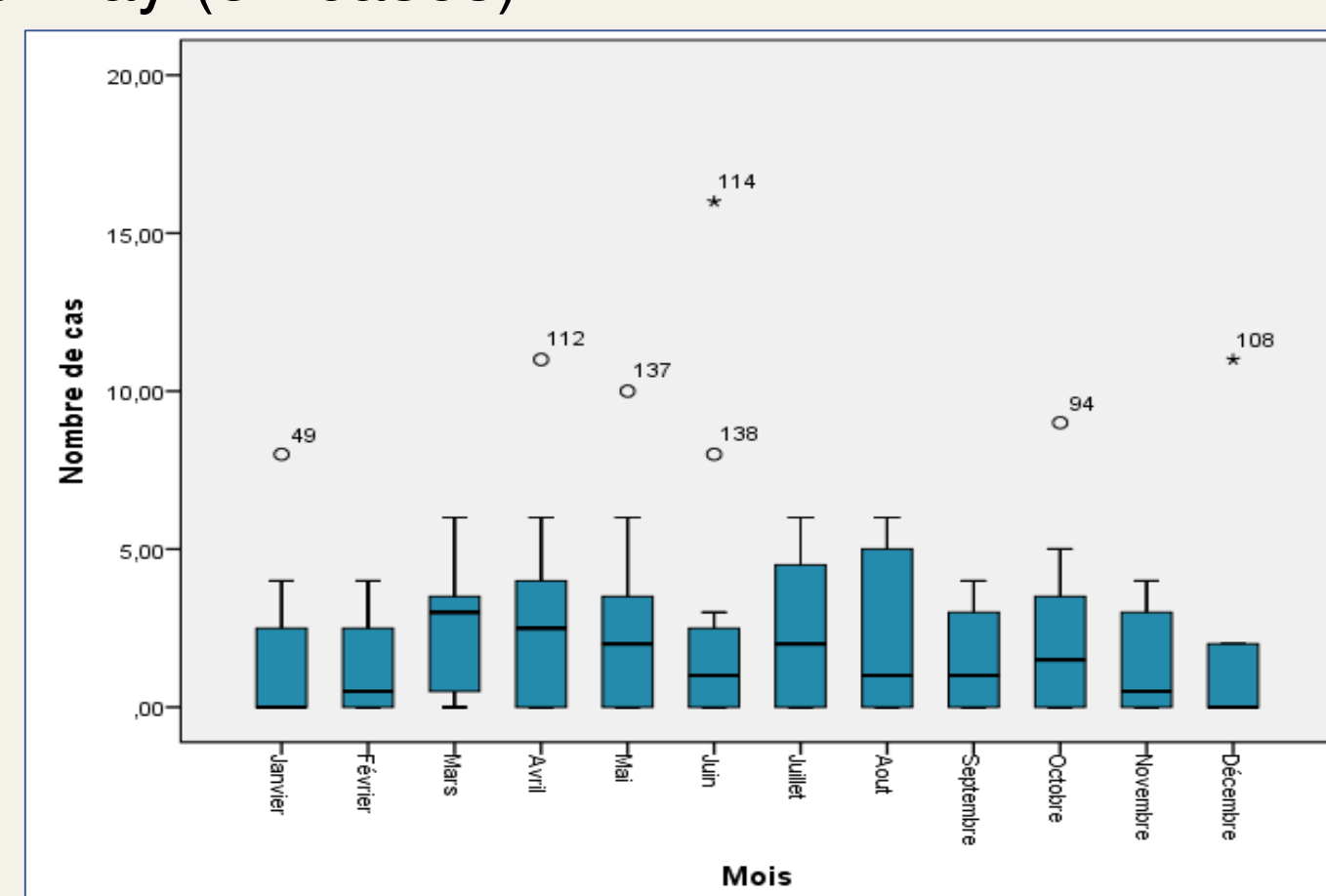


Figure 2: Monthly Distribution of Human Brucellosis Cases (2010–2021)

3- Age and Sex Distribution

- Most affected age group:** 22–44 years, followed by 45–65 years.
- Youngest case:** 2 years old; oldest: 87 years; no cases under 1 year old.
- Average age:** 45 years.
- More frequent in females** (53.72%) than males (46.28%), sex ratio = 0.86.
- Both sexes **most affected** in the 22–44 age range.

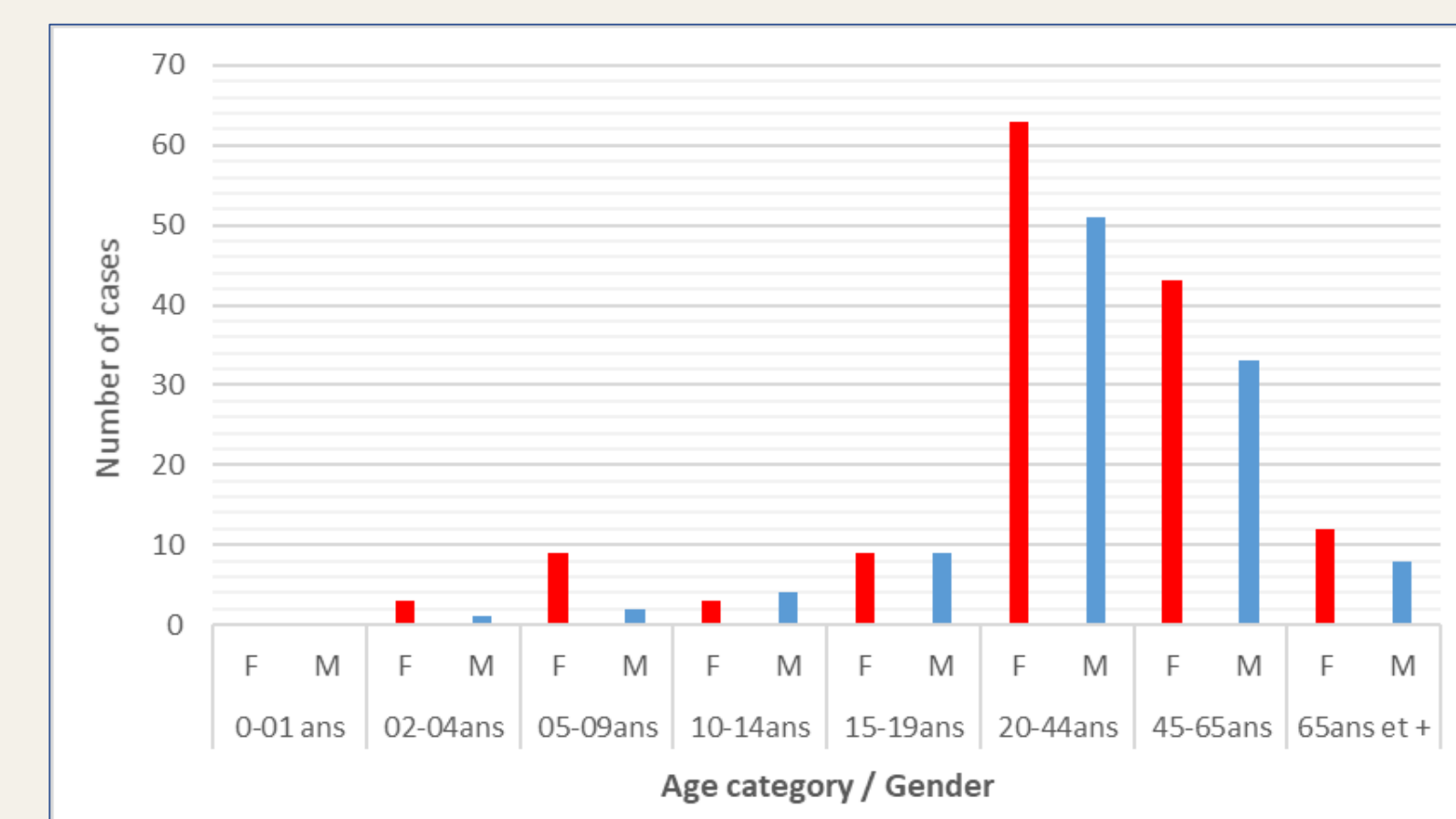


Figure 3: Changes in the Number of Human Brucellosis Cases by Age and Sex (2010–2021)

4- Spatial Distribution

- Cases reported in almost all communes of El Tarf, except Chihani.
- Most affected communes: Bouhadjer, Hamman Béni Salah, Bougous, Oued Zitoun, Bouteldja.
- Least affected: Echatt, Chbaita Moukhtar, El Aioun, Raml Souk.

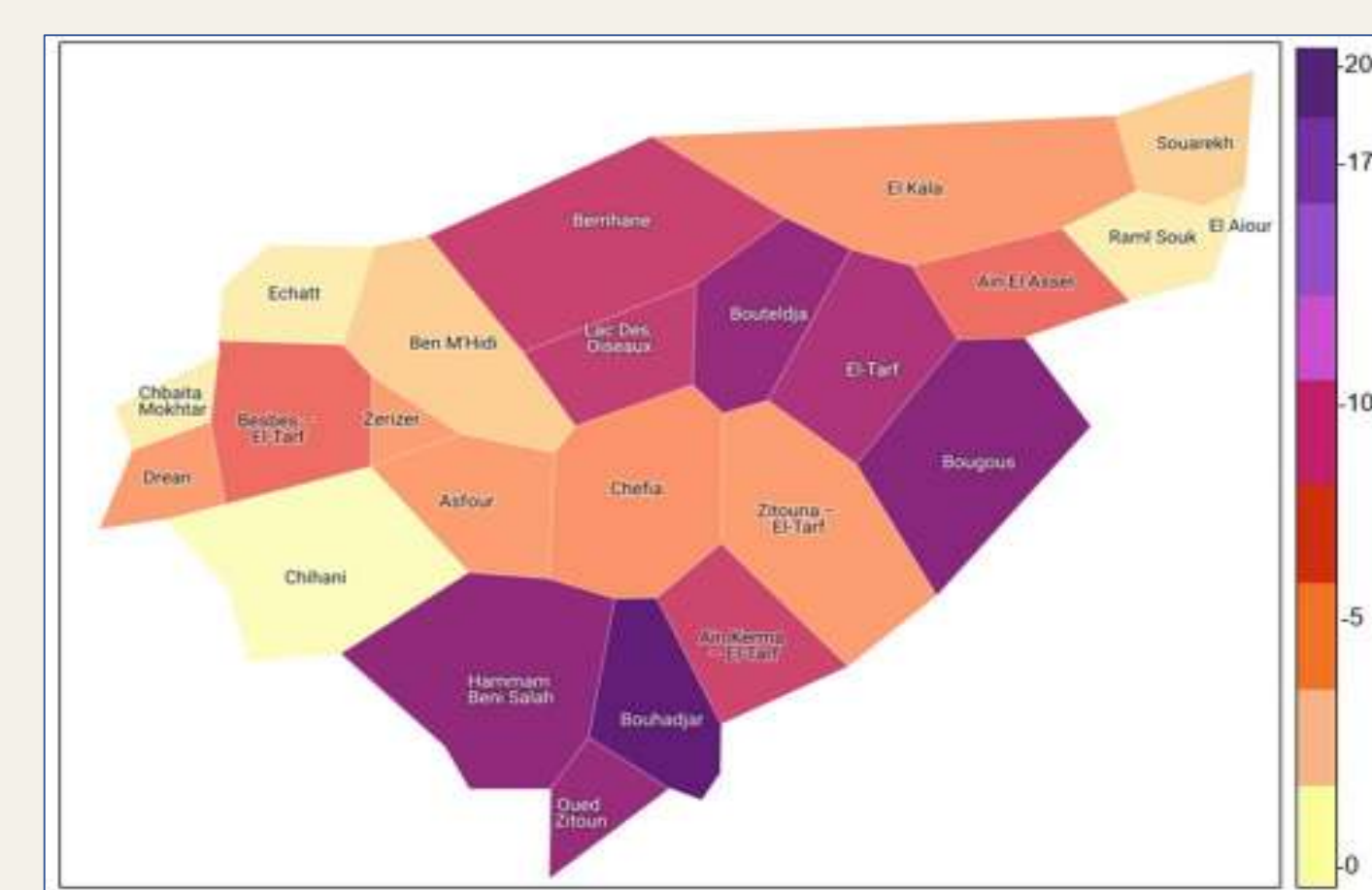


Figure 4: Geographical Distribution of Human Brucellosis in El Tarf Province (2010–2021)

5- Spatio-temporal Distribution

- Epizootic peaks noted in Bouhadjer (2016, 2019, 2021), Hamman Béni Salah (2017), and Bougous (2019).
- Number of affected communes varied yearly from 1 to 14, with no commune consistently affected.

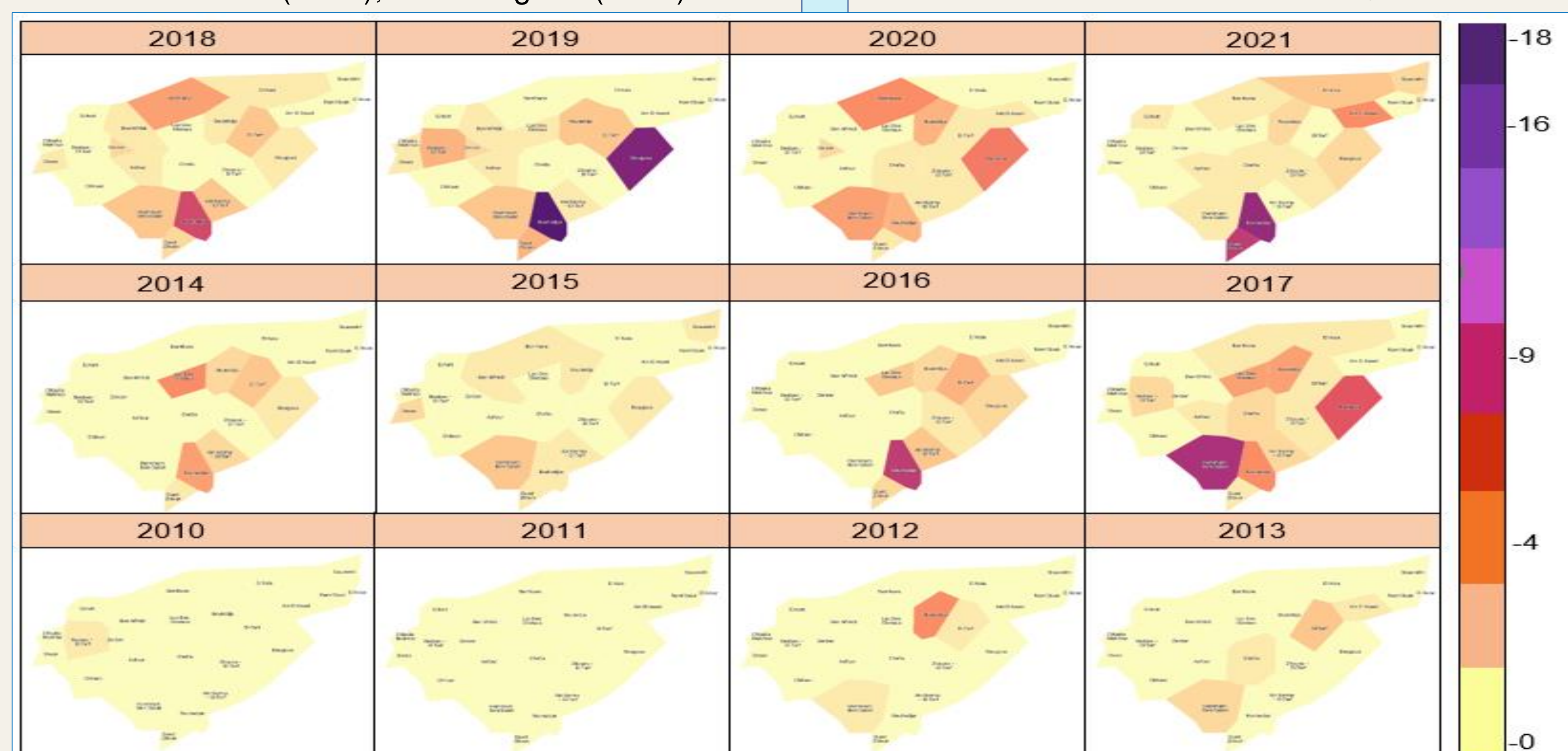


Figure 5: Spatiotemporal Distribution of Human Brucellosis Cases in El Tarf Province (2010–2021)

Conclusion

The current surveillance system relies solely on passive case detection, likely leading to underreporting of human brucellosis cases. This highlights the need for complementary active surveillance to better capture true incidence and improve control. Despite state measures, brucellosis remains endemic in El Tarf with variable incidence. The lack of livestock identification and tracking also limits surveillance effectiveness. Passive surveillance alone is insufficient; active monitoring and regular system evaluation are essential. Further studies are needed to understand the disease's persistence despite public health efforts.